



MID SEMESTER EXAMINATION, AUTUMN-2024
School of Applied Sciences, KIIT-DU

B. Tech.
 1st Semester (2024AB)
 Autumn 2024 (SAS)

Course: Chemistry

Code: CH10001

Full Marks: 20

Time: 90 minutes

Answer any FOUR QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

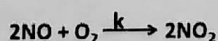
Q.1		Marks	CO
a)	Predict the nature of the line (slope) by using Van't Hoff isochore relation when you plot $\ln K_{eq}$ vs. $1/T$ for exothermic and endothermic reaction.	1	1
b)	Calculate ΔG for isothermal reversible compression of a mole of an ideal gas from 1 atm to 50 atm at 303 K.	1	1
c)	Show that the entropy change during the reversible isothermal process is constant.	1	1
d)	At which condition, the process will be spontaneous at all temperature	1	1
	i) $\Delta H > 0$ and $\Delta S < 0$ ii) $\Delta H < 0$ and $\Delta S > 0$ iii) $\Delta H < 0$ and $\Delta S < 0$ iv) $\Delta H > 0$ and $\Delta S > 0$		
e)	The decomposition of a compound follows first order kinetics. The Rate constant of the reaction is 0.075 s^{-1} . Find the half-life of the compound.	1	2

Q.2		Marks	CO
a)	Show that during the mixing of two ideal gases the ΔS will be always positive	3	1
b)	At what temperature will water boil when the atmospheric pressure is 600 mm of Hg? Given latent heat of vaporization is 545.5 cal/g.	2	1

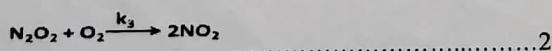
Q.3		Marks	CO
a)	Derive the integrated form of Clausius-Clapeyron equation for $\text{Water (liquid)} \rightleftharpoons \text{Water (vapor)}$	3	1
b)	For a reaction, the value of enthalpy change and free energy change are -68.32 and -56.69 kcal respectively at 25 °C. Calculate the value of free energy change at 30 °C.	2	1

Q.4		Marks	CO
a)	For a hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below, $A_2 \rightleftharpoons A + A$(Fast) $A + B_2 \rightarrow AB + B$(Slow) $A + B \rightarrow AB$(Fast) Find the overall order of the reaction.	3	2
b)	The Equilibrium constant (K_p) for a reaction at 327 °C and 427 °C are 10^{-12} and 10^{-7} respectively. Calculate the enthalpy of the reaction.	2	1

Q.5		Marks	CO
a)	For the following reaction	3	2



the proposed reaction mechanism is;



Assuming all the reactions are elementary, find the relation between overall rate constant k with k_1 , k_2 and k_3 .

b)	A chemical process has $\Delta H = 400 \text{ J mol}^{-1}$ and $\Delta S = 80 \text{ J K}^{-1} \text{ mol}^{-1}$, predict the minimum temperature to make this process spontaneous.	2	1
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